LESSON 7.1

Explore Integers and Their Opposites

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use integers to represent quantities in everyday life, explain what 0 means in each situation, and name the opposite of an integer on the number line.

WHY THIS MATTERS The lowest elevation in North America is Badwater Basin in Death Valley: 86 meters below sea level. When we measure elevation, sea level is defined as 0. So 86 meters below sea level cannot be written as plain 86 — that would put it ABOVE the water. Numbers less than zero are negative numbers, written with a − sign, so the elevation of Badwater Basin is −86 meters. On a number line, −86 is 86 units to the LEFT of 0, while 86 is 86 units to the RIGHT of 0. Those two numbers are opposites.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Integer *Spanish: Número entero*	The counting numbers, their opposites, and 0.	−9, −1, 0, 6, 230
Opposite *Spanish: Opuesto*	A number that is the same distance from 0 as another number, but on the other side of 0.	−6 and 6 are opposites
Negative number *Spanish: Número negativo*	A number less than 0, written with a − sign. On a number line it sits to the left of 0.	−9 is read negative nine
Positive number *Spanish: Número positivo*	A number greater than 0. On a number line it sits to the right of 0.	6, 19, 230
Number line *Spanish: Recta numérica*	A straight line where numbers are placed in order. It extends to the left of 0 to show values less than 0.	left is less, right is more
Opposite of the opposite *Spanish: Opuesto del opuesto*	Taking the opposite twice brings you back to the first number.	−(−6) = 6

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A game score starts at 0. You earn 8 points, then you lose 8 points. Write each as an integer.

1. 0 means the starting score.
2. Earn 8 points: 8. Lose 8 points: −8.
3. Both are 8 units from 0, on opposite sides.

Answer: 8 and −8 are opposites

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

One day in January the temperature in Toronto was 9°C and in Reykjavik it was −9°C.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The integer _____ represents _____ .
- Zero stands for _____ in this situation.
- The opposite of _____ is _____ because _____ .
- On the number line, _____ sits _____ units to the _____ of 0.

Word bank integer · opposite · negative · positive · number line · zero · distance · above · below · left · right · sign

WATCH OUT!

A common mistake

The most common mistake in this lesson is thinking that the opposite of a negative integer is still negative

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Write the integer that represents each situation. Remember to decide what 0 means first.

Situation	Integer
A football play with a 12-yard loss	
A hiker 250 meters above sea level	
7 degrees below zero	
A submarine resting at the surface of the water	

STEPS TO TRY

1. Each row describes a quantity measured from a starting point. That starting point is 0.
2. Words like 'loss' and 'below' put the quantity under 0; words like 'above' put it over 0.
3. Row 1 measures from the line of scrimmage, and a loss moves backward from it — so the integer needs a negative sign.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Badwater Basin is 86 meters below sea level, and sea level is 0. Which integer represents that elevation?

(A) -86

(B) -68

(C) 0

(D) 86

STEPS TO TRY

1. Sea level is 0. Is 'below sea level' more than 0 or less than 0?
2. Numbers less than 0 are negative numbers and are written with a − sign in front.
3. Keep the distance the sign gives you — 86 meters — and add the symbol that shows the point is below 0.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

In Reykjavik the temperature was -9°C. What is the opposite of -9?

(A) -18
(B) -9
(C) 0
(D) 9

STEPS TO TRY

1. 'Opposite' means the same distance from 0, but on the other side of 0.
2. -9 sits 9 units to the LEFT of 0. Count 9 units to the RIGHT of 0 instead.
3. Keep the 9 exactly the same and change only the direction, which means changing only the sign.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the opposite of 0?

(A) A number written -0, which is less than 0
(B) Every negative number
(C) Zero has no opposite
(D) 0

STEPS TO TRY

1. The opposite of a number is the same distance from 0 on the other side.
2. Ask yourself: how many units away from 0 is 0 itself?
3. If a number is 0 units from 0, moving to the other side takes you to the very same point.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide what 0 meant in each situation, and how did that decide the sign of the integer?

SENTENCE STARTER In this situation 0 means ___. The quantity is ___ zero, so I represent it with a ___ integer, which is ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help